

Coordinate Systems

Coordinate Systems

Defn. If B is a basis of $\mathbb{R}^n$, then $[x]_B$ gives the coefficients used to express x as a linear combination of vectors in B . It is called the **coordinates** of x relative to B .

For example, If B is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$ and $x = (5, -6)$, then $[x]_B = (7, -2)$. (Check: calculate $7 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.)

Change of Coordinates

Defn. In $\mathbb{R}^n$, the **change-of-coordinates matrix** P_B has B as its columns.

Thus

$$x = P_B [x]_B$$

where x is given in the standard basis.

Change of Coordinates Again

Defn. For bases B and C , the **change-of-coordinates** matrix $P_{C \leftarrow B}$ is such that

$$[x]_C = P_{C \leftarrow B} [x]_B$$

The columns of $P_{C \leftarrow B}$ express each vector of B in terms of C . Also $P_{C \leftarrow B} = P_C^{-1} P_B$.

Summary

If B is a basis, then $[x]_B$ gives the coefficients to express x as a linear combination of B .

In $\mathbb{R}^n$ the change-of-coordinates matrix P_B has B as its columns and $x = P_B [x]_B$ for x in the standard basis. For two bases, the change-of-coordinates matrix expresses each vector of one in terms of the other.